# EDA01: App Insights Analysis - Solution Guide

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**Data Source:** Google Play Store (10k records)

**Section 1: Basic Insights (10 Questions)**

1. **Total number of apps in the dataset?**
    - **Answer:** 7,016 Apps.
    - **Logic:** Total Apps = COUNT('App Data'[App])
2. **What is the average user rating across the store?**
    - **Answer:** 4.17
    - **Logic:** Average Rating = AVERAGE('App Data'[Rating])
3. **Total number of reviews generated by all apps?**
    - **Answer:** 2 Billion (2,000,000,000).
    - **Logic:** Total Reviews = SUM('App Data'[Reviews])
4. **Total number of installs across all categories?**
    - **Answer:** 58 Billion (58,000,000,000).
    - **Logic:** Total Installs = SUM('App Data'[Installs])
5. **Which category has the highest number of apps?**
    - **Answer: Family** (1,591 apps).
    - **Logic:** Clustered Bar Chart of Count of App by Category.
6. **What is the distribution of Free vs. Paid apps?**
    - **Answer: Free: 92.63%** (6,882) | **Paid: 7.37%** (548).
    - **Logic:** Donut Chart using Type and Count of App.
7. **What is the most common Content Rating?**
    - **Answer: Everyone** (80.25%).
    - **Logic:** Treemap of Content Rating.

8. **Which category has the highest average app price?**
    - **Answer: Finance** ($9.28 average).
    - **Logic:** Average Price = AVERAGE('App Data'[Price]) grouped by Category.
9. **How many apps have a "Mature 17+" rating?**
    - **Answer:** 4.47% of the dataset (~313 apps).
    - **Logic:** Calculated via Content Rating distribution chart.
10. **What is the 2nd most popular category by app count?**
    - **Answer: Game** (959 apps).

**Section 2: Medium Insights (10 Questions)**

11. **Which category has the highest average rating?**
    - **Answer: Events** (4.48).
    - **Logic:** Bar Chart sorted by Average Rating.
12. **Which category has the most installs?**
    - **Answer: Game** (Approx. 30 Billion).
    - **Logic:** Bubble Chart (Size = Total Installs).
13. **What is the average rating for "Mass Market" apps (Installs > 1M)?**
    - **Answer:** 4.3
    - **Logic:** Avg Rating by Installs Group using a conditional column for grouping.
14. **What is the average rating for "Niche" apps?**
    - **Answer:** 4.1
15. **In which year did the app market see the biggest spike in updates/releases?**
    - **Answer: 2018** (4,591 apps).
    - **Logic:** Line Chart: Year vs. Count of App.
16. **What was the growth in app releases between 2017 and 2018?**
    - **Answer:** Growth from 793 apps to 4,591 apps (approx. 479% increase).

17. **Which is the 2nd most expensive category on average?**
    - **Answer: Lifestyle** ($7.15).
18. **Top 3 categories by total installs?**
    - **Answer:** 1. Game, 2. Communication, 3. Tools.
19. **Does the "Teen" content rating have a higher avg rating than "Everyone"?**
    - **Answer:** No, "Everyone" generally maintains a higher rating density.
20. **Is there a relationship between Reviews and Ratings?**
    - **Answer:** Yes, higher review volumes are clustered around the 4.0 - 4.5 rating range.

**Section 3: Advanced Insights (5 Questions)**

21. **What is the "Sweet Spot" for app size to maximize user rating?**
    - **Answer: 80MB - 100MB**.
    - **Logic:** Line chart shows rating peaks at ~4.30 for this size bin.
22. **Do Paid apps have higher satisfaction than Free apps?**
    - **Answer: Yes.** Paid apps average ~4.25 while Free apps average ~4.17.
    - **Logic:** Bar chart comparing Avg Rating by Type.
23. **How does Price impact engagement (Reviews)?**
    - **Answer:** Negatively. Free apps account for 2B reviews; Paid apps have statistically negligible review volume.
24. **Identify a "High-Value Niche" category (Low volume, High rating).**
    - **Answer: Events and Education**. (Highest ratings despite lower total installs).
25. **Strategic recommendation for a new developer?**
    - **Answer:** Target the **Game** category for volume, but aim for an **~80MB** build size to ensure a "premium" quality perception.